

Ex. No 4:- Analyze email headers and detect email spoofing using MHA

Date:-

Aim:-

To analyze an email header using MXToolbox's Mail Header Analyzer and check its authentication results to detect email spoofing.

Description:-

Here's a step-by-step guide on how to execute and analyze an email header:

1. Access the Email Header

- **Gmail:** Open the email, click the three dots (More) in the upper right corner, and select "Show original."
- **Outlook:** Open the email, click on "File," then "Properties," and find the "Internet headers" box.
- **Yahoo:** Open the email, click on the three dots (More), and select "View raw message."

2. Copy the Email Header

- Once you access the header, copy all the text that appears. This text contains various metadata fields about the email's journey from the sender to your inbox.

3. Identify Key Header Fields

- **From:** The sender's email address.
- **To:** The recipient's email address.
- **Date:** The date and time the email was sent.
- **Subject:** The email's subject line.
- **Return-Path:** The return address if the email bounces.
- **Received:** A list of servers the email passed through on its way to the recipient. This is usually the most critical part of header analysis.
- **Message-ID:** A unique identifier for the email.
- **SPF (Sender Policy Framework) / DKIM (DomainKeys Identified Mail):** Indicators of email authentication.

4. Analyze the 'Received' Fields

- The Received fields show the path the email took from the sender to the recipient. These fields are listed in reverse order (from the last server to the first).
- Each Received line will show:

- **The sending server's hostname/IP address.** ○
- The receiving server's hostname/IP address.**
- **The date and time the email passed through.**

5. Check for IP Addresses and Hostnames

- Use tools like **WHOIS** or online IP lookup services to identify the geographical location and ownership of the IP addresses found in the Received lines.
- Check if any IP addresses are suspicious or if the hostname does not match the expected sending server.

6. Examine the SPF, DKIM, and DMARC Results

- **SPF:** Checks if the sender's IP is authorized to send emails for the domain.
 - **Pass:** Indicates the IP is authorized.
 - **Fail:** The IP is not authorized and could be a sign of spoofing.
- **DKIM:** Verifies that the email content has not been altered in transit.
 - **Pass:** The email has not been tampered with.
 - **Fail:** The email content may have been altered.
- **DMARC:** A policy that uses SPF and DKIM to authenticate the email.
 - **Pass:** Indicates alignment with SPF/DKIM policies.
 - **Fail:** Indicates possible spoofing or tampering.

7. Analyze the 'Message-ID'

- The Message-ID is unique to each email. It usually includes the domain of the sending server.
- Check if the domain in the Message-ID matches the sender's domain. If not, it could be a sign of spoofing.

8. Look for Anomalies

- **Mismatch in Email Domains:** Check if the From domain matches the domains in ReturnPath, Message-ID, and other authentication results.
- **Strange IP Addresses:** Be wary of IP addresses that do not match the known sending server or are from unexpected locations.
- **Suspicious Time Stamps:** Check for significant time discrepancies in the Received lines, which could indicate delays or rerouting.

9. Use Online Tools

- Use online email header analysis tools like **MXToolbox** or **Google's G Suite Toolbox** to automatically parse and analyze the email header.

10. Document and Report Findings

- Document all your findings, especially if you detect phishing or spoofing.

- Report suspicious emails to your IT department or email provider for further investigation.

Example of a Header Analysis

Here's a quick example with a sample header:

vbnet Copy

code

Received: from mail.example.com (mail.example.com [192.0.2.1]) by mail.receiver.com with ESMTP id u29si8604336pjs.40.2023.08.10.07.00.16;

Thu, 10 Aug 2023 07:00:16 -0700 (PDT)

Received: by mail.example.com with SMTP id a1mr1243772ywh.51;

Thu, 10 Aug 2023 07:00:15 -0700 (PDT)

Message-ID: <CA+7eu=4pSeXgQ@mail.example.com>

- **Check IP Addresses:** Lookup 192.0.2.1 and ensure it is associated with mail.example.com.
- **Verify SPF/DKIM:** Check if mail.example.com is authorized to send emails for this domain.
- **Compare Time Stamps:** Ensure the times make sense chronologically without unusual delays.

Screen Shots:

Original message

Message ID	<jBeaMyITNm1zcTUu9o2hg@geopod-ismtpd-21>
Created on:	14 August 2026 at 22:02 (Delivered after 1 second)
From:	Flightradar24 <newsletter@fr24.com>
To:	nikhiljangama18@gmail.com
Subject:	There are fewer airship pilots than astronauts
SPF:	PASS with IP 167.89.98.228 Learn more
DKIM:	'PASS' with domain fr24.com Learn more
DMARC:	'PASS' Learn more

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MX TOOLBOX
SUPERTOOL

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Email Header Analyzer

Paste Header:

```
pDX3Elv-2Bso3DWhNMNsV-2FjCfBgFbPCU1sGY2quncSAPEcpBVDQCAgDwwPRjYVdj8QcVe61=
wMT3dhwmX0abJh6XyvjmkMfGxwJGkx370EC0Lc5v6QrC8jTWRjZT-2FJD-2FOXs3gC7nF0D-2Bw=
xKwvZRhHXqDMsklbZl8-2BBrKvfeFn2OA-3D-3D" alt="3D" width="3D" height="3D" style="border="3D" style="border="3D" height: 1px !important; width: 1px !important; border-width:
th: 0 !important; margin-top: 0 !important; margin-bottom: 0 !important; margin-left:
right: 0 !important; margin-left: 0 !important; padding-top: 0 !important; padding=
-bottom: 0 !important; padding-right: 0 !important; padding-left: 0 !important; "="
/></body>
</html>
--02da8a98a7d9cb21a3b60549f150966f94dcaae624480fc4dde979a091--
```

Analyze Header

ABOUT EMAIL HEADERS

This tool will make email headers human readable by parsing them according to RFC 822. Email headers are present on every email you receive via the Internet and can provide valuable diagnostic information like hop delays, anti-spam results and more. If you need help getting copies of your email headers, [just read this tutorial](#).

Headers Found

Header Name	Header Value
Delivered-To	nikhiljangama18@gmail.com
X-Received	by 2002:a05:6a20:7fa9:b0:3cc:22bb:8911 with SMTP id adf61e73a8af0-3cc71ac37f9mr8429098637.6.1786725133451; Fri, 14 Aug 2026 09:32:13 -0700 (PDT)
ARC-Seal	i=1; a=rsa-sha256; t=1786725133; cv=none; d=google.com; s=arc-20260327; b=Ht0t4VVzTFu6+bREWUSAaMFrJph+b7+U964/E2yE7fogmt7nPousBX/3dTlWuWNeI uwP1g1cehnPLaTlqmpLGo/G2ledilKODGIsX2YclhAKHKEqMPq7ifvRJTGilxbq7H9 pLo8jddd9lv0cmMj37GZLwaCOG8GRdjBORsWIdb138qDQXFeUU95wRYANhY3Fv6Djn0 J7R7CuEgh64b0HVuTAHbZ1kWeDWNb2YKvjbyPn4bOi9cEHK7lIII83VwcXI8C6x52EpC KNuqEPWBBAGDijMA8mK8BDfyxklrHMHl9wblYWWRFobJdsMq/HvNa/wEDUcnTpg01Znp vV5Q==
ARC-Message-Signature	i=1; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20260327; h=list-unsubscribe-post:list-unsubscribe-to:reply-to:message-id:subject:mime-version:from:date:dkim-signature:dkim-signature; bh=WWJT2AMda1oUmhmyuHcYyJmCcD/aqMLxHxU4Y4ZsU=; fh=L9xcH35JYkLV1ZiPv9LlOGk412kHsS0Cz3nw0puK30=; b=cwfw/S9G5eqDpFaIGrF1+anGAs6kmwTortro9fLSE8r5x4LzkjokHQdzl++xg2qS+ SCUHpyVJCHW4i/xph09RxMHChVhIVPkGtvtl+W/FJygGII0ifSQ5Hr+wSRXG1J8+PkJ d02n7xopwo8aVOSkqW6enO/rwe5lvmkbKc6OZbqkQCZNSk816G49FmPNvgcp3zfl8w3X NXbjx80pCuiReF2UENpc3Q1VU06vzg7SMQN3pHk9Z12i5EifxEDBRi840R7JDB9qajXc TrZ253iCdcOZZy5eeuwxm9v8JkuLwK63FjWYbDyr2n2xk6qD2Uk4cOzDfUki7IH7 06BA==; dara=google.com
ARC-Authentication-Results	i=1; mx.google.com; dkim=pass header.i=@fr24.com header.s=s1 header.b=eA3coyzY; dkim=pass header.i=@sendgrid.info header.s=smtapi header.b="Laiicm7/"; spf=pass (google.com: domain of bounces+7728411-e93a-nikhiljangama18@gmail.com@post.fr24.com designates 167.89.98.228 as permitted sender) smtp.mailfrom="bounces+7728411-e93a-nikhiljangama18@gmail.com@post.fr24.com"; dmarc=pass (p=QUARANTINE sp=QUARANTINE dis=NONE) header.from=fr24.com
Return-Path	<bounces+7728411-e93a-nikhiljangama18@gmail.com@post.fr24.com>
Received-SPF	pass (google.com: domain of bounces+7728411-e93a-nikhiljangama18@gmail.com@post.fr24.com designates 167.89.98.228 as permitted sender) client-ip=167.89.98.228;
Authentication-Results	mx.google.com; dkim=pass header.i=@fr24.com header.s=s1 header.b=eA3coyzY; dkim=pass header.i=@sendgrid.info header.s=smtapi header.b="Laiicm7/"; spf=pass (google.com: domain of bounces+7728411-e93a-nikhiljangama18@gmail.com@post.fr24.com designates 167.89.98.228 as permitted sender) smtp.mailfrom="bounces+7728411-e93a-nikhiljangama18@gmail.com@post.fr24.com"; dmarc=pass (p=QUARANTINE sp=QUARANTINE dis=NONE) header.from=fr24.com
DKIM-Signature	v=1; a=rsa-sha256; c=relaxed/relaxed; d=fr24.com; h=content-type:date:from:mime-version:subject:reply-to:to: list-unsubscribe:list-unsubscribe-post:cc:content-type:date:feed-back-id:from:subject:to; s=s1; t=1786725132; bh=WWJT2AMda1oUmhmyuHcYyJmCcD/aqMLxHxU4Y4ZsU=; b=eA3coyzY5UPi4EeDYlaeB/jpAEAfzcHN3dafNzJkNaeyRvJiEOwJ+2WdCzocj9Z2Uv adMx09e8RssGAVdE7gyr1CsavyMOhOom7lrVvK0PLCkgENM9BilLzVm8c/ZxIP88pKqIE GVknoKjhT1n6FWge9AELLB1Umsk/BHrqJOviGbbKBV/WvjdfVGUXBk+OLknvm7o4pAG hmvAVoANL5MhoMnQZvVMFR6nKXt6mBGztjDV9PVu3pknoFeDOiLsiKO715D093f4KzrF eAV16AtjKsm6jAFmOFxQ4rPgk9Zjfn5oFtyuF6bZsdJQbXC+0mBP/QK9NxrKA==

Received Header

Delivered-To: nikhiljangama18@gmail.com
Received: by 2002:a17:907:d310:b0:c20:eddb:4981 with SMTP id vg16csp2346230ejc; Fri, 14 Aug 2026 09:32:13 -0700 (PDT)
X-Received: by 2002:a05:6a20:7fa9:b0:3cc:22bb:8911 with SMTP id adf61e73a8af0-3cc71ac37f9mr8429098637.6.1786725133451; Fri, 14 Aug 2026 09:32:13 -0700 (PDT)
ARC-Seal: i=1; a=rsa-sha256; t=1786725133; cv=none; d=google.com; s=arc-20260327; b=Ht0t4VVzTFu6+bREWUSAaMFrJph+b7+U964/E2yE7fogmt7nPousBX/3dTlWuWNeI uwP1g1cehnPLaTlqmpLGo/G2ledilKODGIsX2YclhAKHKEqMPq7ifvRJTGilxbq7H9 pLo8jddd9lv0cmMj37GZLwaCOG8GRdjBORsWIdb138qDQXFeUU95wRYANhY3Fv6Djn0 J7R7CuEgh64b0HVuTAHbZ1kWeDWNb2YKvjbyPn4bOi9cEHK7lIII83VwcXI8C6x52EpC KNuqEPWBBAGDijMA8mK8BDfyxklrHMHl9wblYWWRFobJdsMq/HvNa/wEDUcnTpg01Znp vV5Q==
ARC-Message-Signature: i=1; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20260327; h=list-unsubscribe-post:list-unsubscribe-to:reply-to:message-id:subject:mime-version:from:date:dkim-signature:dkim-signature; bh=WWJT2AMda1oUmhmyuHcYyJmCcD/aqMLxHxU4Y4ZsU=; fh=L9xcH35JYkLV1ZiPv9LlOGk412kHsS0Cz3nw0puK30=; b=cwfw/S9G5eqDpFaIGrF1+anGAs6kmwTortro9fLSE8r5x4LzkjokHQdzl++xg2qS+ SCUHpyVJCHW4i/xph09RxMHChVhIVPkGtvtl+W/FJygGII0ifSQ5Hr+wSRXG1J8+PkJ d02n7xopwo8aVOSkqW6enO/rwe5lvmkbKc6OZbqkQCZNSk816G49FmPNvgcp3zfl8w3X NXbjx80pCuiReF2UENpc3Q1VU06vzg7SMQN3pHk9Z12i5EifxEDBRi840R7JDB9qajXc TrZ253iCdcOZZy5eeuwxm9v8JkuLwK63FjWYbDyr2n2xk6qD2Uk4cOzDfUki7IH7 06BA==; dara=google.com
Return-Path: <bounces+7728411-e93a-nikhiljangama18@gmail.com@post.fr24.com>
Received: from o167-89-98-228.outbound-mail.sendgrid.net (o167-89-98-228.outbound-mail.sendgrid.net. [167.89.98.228]) by mx.google.com with ESMTPS id 41be03b00d21.
Received-SPF: pass (google.com: domain of bounces+7728411-e93a-nikhiljangama18@gmail.com@post.fr24.com designates 167.89.98.228 as permitted sender) client-ip=167.89.98.228; envelope-from=bounces+7728411-e93a-nikhiljangama18@gmail.com@post.fr24.com; dkim=pass header.i=@fr24.com header.s=s1 header.b=eA3coyzY; dkim=pass header.i=@sendgrid.info header.s=smtapi header.b="Laiicm7/"; spf=pass (google.com: domain of bounces+7728411-e93a-nikhiljangama18@gmail.com@post.fr24.com designates 167.89.98.228 as permitted sender) smtp.mailfrom="bounces+7728411-e93a-nikhiljangama18@gmail.com@post.fr24.com"; dmarc=pass (p=QUARANTINE sp=QUARANTINE dis=NONE) header.from=fr24.com
Content-Type: multipart/alternative; boundary=02da8a98a7d9cb21a3b60549f150966f94dcaae624480fc4dde979a091
Date: Fri, 14 Aug 2026 16:32:12 +0000 (UTC)
From: Flightradar24 <newsletter@fr24.com>
Mime-Version: 1.0
Subject: There are fewer airship pilots than astronauts
Message-ID: <-jBeaMyLTNm1zcTUu9o2hg@geopod-ismtpd-21>
Reply-To: newsletter@fr24.com
X-SG-EID: =?us-ascii?Q?u001=2ETILndKXtTPodbnOVpGTegD1sp35NTBBK1r3WqWav2fz4wsPp0liyF+FY=2F?= =?us-ascii?Q?Cet+xlLdtWggcah=2FDFgY=2FDOIhT2dRK1LDmbwXRF?= =?us-ascii?Q?u001=2E3RPXmXhVfbZY57ufbcgX07Arz4f8=2Fa3+z0BwdJS1d60G5tc5jdOHeruL1?= =?us-ascii?Q?Y8mXdbS7mqilthyHAREXG6oscc5f1mBC7VbkFN?= =?us-ascii?Q?1C:To: nikhiljangama18@gmail.com
X-Entity-ID: u001.iy+wlS3W+amKFWXUVpQ0mw==
List-Unsubscribe: =?us-ascii?Q?3chttps=3A=2F=2Fu7728411=2Ect=2F=2Fsendgrid=2Enet=2Flu=2F=2Funsunsubscribe=3Foc=3Du001=2E5j8ph?= =?us-ascii?Q?HG5xvXotJu2fc0cHHSUL587cl21vList-Unsubscribe-Post: List-Unsubscribe=One-Click
--02da8a98a7d9cb21a3b60549f150966f94dcaae624480fc4dde979a091
Content-Transfer-Encoding: quoted-printable

Rubrics:-

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result:-

The header analysis showed that SPF, DKIM, and DMARC all passed, with the sending IP matching the authorized sender's domain and no delivery delay. This confirmed the email was genuine and not spoofed.